

Отчёт к лабораторной работе №3 по предмету «Администрирование в ОС Linux» «Установка программного обеспечения»

- 1) Выведен список подключённых АРТ-репозиториев. Установлено, что в системе Ubuntu используется формат *.sources, отображены параметры для основных и security-репозиториев

```
[16-03-2026 18:04:07] {40ab33fa} student@vm-amd:~$ cat /etc/apt/sources.list
# Ubuntu sources have moved to /etc/apt/sources.list.d/ubuntu.sources
[16-03-2026 18:04:45] {40ab33fa} student@vm-amd:~$ cat /etc/apt/sources.list
ls -l /etc/apt/sources.list.d/
# Ubuntu sources have moved to /etc/apt/sources.list.d/ubuntu.sources
ubuntu.sources
ubuntu.sources.curtin.orig
[16-03-2026 18:04:58] {40ab33fa} student@vm-amd:~$ grep -R --no-filename -E '^(deb|deb-src)' /etc/apt/sources.list /etc/apt/sources.list.d/*.list 2>/dev/null
[16-03-2026 18:05:09] {40ab33fa} student@vm-amd:~$ grep -R --no-filename -E '^(Types|URIs|Suites|Components):' /etc/apt/sources.list.d/*.sources 2>/dev/null
Types: deb
URIs: http://archive.ubuntu.com/ubuntu/
Suites: noble noble-updates noble-backports
Components: main restricted universe multiverse
Types: deb
URIs: http://security.ubuntu.com/ubuntu/
Suites: noble-security
Components: main restricted universe multiverse
```

- 2) Выполнено обновление локальных индексов пакетов командой `sudo apt update`. Индексы успешно получены из подключённых репозиториев, также отображена информация о доступных обновлениях пакетов

```
[16-03-2026 18:04:15] {40ab33fa} student@vm-amd:~$ sudo apt update
[sudo] password for student:
Get:1 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:2 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1,538 kB]
Get:3 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [246 kB]
Get:4 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.5 kB]
Get:5 http://security.ubuntu.com/ubuntu noble-security/main amd64 c-n-f Metadata [10.5 kB]
Get:6 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [2,677 kB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/restricted Translation-en [622 kB]
Get:8 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Components [212 B]
Get:9 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 c-n-f Metadata [544 B]
Get:10 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [976 kB]
Get:11 http://security.ubuntu.com/ubuntu noble-security/universe Translation-en [249 kB]
Get:12 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [74.2 kB]
Get:13 http://security.ubuntu.com/ubuntu noble-security/universe amd64 c-n-f Metadata [20.8 kB]
Get:14 http://security.ubuntu.com/ubuntu noble-security/multiverse Translation-en [6,732 B]
Get:15 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Components [212 B]
Hit:16 http://archive.ubuntu.com/ubuntu noble InRelease
Get:17 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:18 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:19 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1,837 kB]
Get:20 http://archive.ubuntu.com/ubuntu noble-updates/main Translation-en [336 kB]
Get:21 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [177 kB]
Get:22 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 c-n-f Metadata [16.7 kB]
Get:23 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [2,819 kB]
Get:24 http://archive.ubuntu.com/ubuntu noble-updates/restricted Translation-en [652 kB]
Get:25 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]
Get:26 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 c-n-f Metadata [556 B]
Get:27 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1,566 kB]
Get:28 http://archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [319 kB]
Get:29 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [386 kB]
Get:30 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 c-n-f Metadata [33.0 kB]
Get:31 http://archive.ubuntu.com/ubuntu noble-updates/multiverse Translation-en [7,044 B]
Get:32 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:33 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Packages [40.4 kB]
Get:34 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [7,348 B]
Get:35 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 c-n-f Metadata [368 B]
Get:36 http://archive.ubuntu.com/ubuntu noble-backports/restricted amd64 Components [216 B]
Get:37 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Packages [30.7 kB]
Get:38 http://archive.ubuntu.com/ubuntu noble-backports/universe Translation-en [18.2 kB]
Get:39 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [13.2 kB]
Get:40 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 c-n-f Metadata [1,488 B]
Get:41 http://archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Packages [780 B]
Get:42 http://archive.ubuntu.com/ubuntu noble-backports/multiverse Translation-en [372 B]
Get:43 http://archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Components [212 B]
Fetched 15.4 MB in 5s (19.2 MB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
123 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

- 3) Выведена информация о метапакете build-essential

```
123 packages can be upgraded. Run 'apt list --upgradable' to see them.
[16-03-2026 18:09:13] {40ab33fa} student@vm-amd:~$ apt show build-essential
Package: build-essential
Version: 12.1ubuntu1
Priority: optional
Build-Essential: yes
Section: devel
Origin: Ubuntu
Maintainer: Ubuntu Developers <ubuntu-devel-discuss@lists.ubuntu.com>
Original-Maintainer: Matthias Klose <dko@debian.org>
Bugs: https://bugs.launchpad.net/ubuntu/+filebug
Installed-Size: 17.4 kB
Depends: libc-dev | libc-dev, gcc (>= 4:12.3), g++ (>= 4:12.3), make, dpkg-dev (>= 1.17.13)
Task: ubuntu-desktop, ubuntu-studio-photography, ubuntu-kylin-desktop, ubuntu-kylin-desktop-minimal, ubuntu-mate-core, ubuntu-mate-desktop, ubuntu-budgie-desktop-minimal, ubuntu-budgie-desktop, ubuntu-budgie-desktop-raspi, ubuntu-unity-desktop, edubuntu-desktop-gnome-minimal, edubuntu-desktop-gnome, edubuntu-desktop-gnome-raspi, ubuntu-cinnamon-desktop-minimal, ubuntu-cinnamon-desktop
Download-Size: 4,928 B
APT-Manual-Installed: yes
APT-Sources: http://archive.ubuntu.com/ubuntu noble/main amd64 Packages
Description: Informational list of build-essential packages
If you do not plan to build Debian packages, you don't need this package. Starting with dpkg (>= 1.14.18) this package is required for building Debian packages.

This package contains an informational list of packages which are considered essential for building Debian packages. This package also depends on the packages on that list, to make it easy to have the build-essential packages installed.

If you have this package installed, you only need to install whatever a package specifies as its build-time dependencies to build the package. Conversely, if you are determining what your package needs to build-depend on, you can always leave out the packages this package depends on.

This package is NOT the definition of what packages are build-essential; the real definition is in the Debian Policy Manual. This package contains merely an informational list, which is all most people need. However, if this package and the manual disagree, the manual is correct.
```

4) Проверена установка метапакета build-essential. Установлено, что пакет уже присутствует в системе в актуальной версии, новых и обновлённых компонентов нет

```
[16-03-2026 18:12:13] {40ab33fa} student@vm-amd:~$ sudo apt install -s build-essential
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
build-essential is already the newest version (12.10ubuntu1).
0 upgraded, 0 newly installed, 0 to remove and 123 not upgraded.
[16-03-2026 18:19:41] {40ab33fa} student@vm-amd:~$ sudo apt install -y build-essential
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
build-essential is already the newest version (12.10ubuntu1).
0 upgraded, 0 newly installed, 0 to remove and 123 not upgraded.
```

5) Выполнен поиск пакета по строке описания, найден пакет bastet

```
0 upgraded, 0 newly installed, 0 to remove and 123 not upgraded.
[16-03-2026 18:19:49] {40ab33fa} student@vm-amd:~$ apt search "clone with a bastard algorithm"
Sorting... Done
Full Text Search... Done
bastet/noble 0.43-7build1 amd64
ncurses Tetris clone with a bastard algorithm
```

6) Создана отдельная директория ~/lab3/src-bastet и выполнена загрузка исходников пакета bastet командой apt source

```
[16-03-2026 18:24:22] {40ab33fa} student@vm-amd:~$ mkdir -p ~/lab3/src-bastet
[16-03-2026 19:13:18] {40ab33fa} student@vm-amd:~$ cd ~/lab3/src-bastet
[16-03-2026 19:13:24] {40ab33fa} student@vm-amd:~/lab3/src-bastet$ apt source bastet
Reading package lists... Done
E: You must put some 'deb-src' URIs in your sources.list
[16-03-2026 19:13:30] {40ab33fa} student@vm-amd:~/lab3/src-bastet$ sudo sed -i 's/^Types: deb$/Types: deb deb-src/' /etc/apt/sources.list.d/ubuntu.sources
[sudo] password for student:
[16-03-2026 19:13:45] {40ab33fa} student@vm-amd:~/lab3/src-bastet$ sudo apt update
Get:1 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Hit:2 http://archive.ubuntu.com/ubuntu noble InRelease
Get:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble/universe Sources [19.8 MB]
Get:6 http://archive.ubuntu.com/ubuntu noble/multiverse Sources [299 kB]
Get:7 http://archive.ubuntu.com/ubuntu noble/main Sources [1,384 kB]
Get:8 http://archive.ubuntu.com/ubuntu noble/restricted Sources [18.7 kB]
Reading package lists... Done
E: Release file for http://security.ubuntu.com/ubuntu/dists/noble-security/InRelease is not valid yet (invalid for another 3h 49min 7s). Updates for this repository will not be applied.
E: Release file for http://archive.ubuntu.com/ubuntu/dists/noble-updates/InRelease is not valid yet (invalid for another 2h 9min 58s). Updates for this repository will not be applied.
E: Release file for http://archive.ubuntu.com/ubuntu/dists/noble-backports/InRelease is not valid yet (invalid for another 2h 12min 50s). Updates for this repository will not be applied.
[16-03-2026 19:14:01] {40ab33fa} student@vm-amd:~/lab3/src-bastet$ apt source bastet
Reading package lists... Done
NOTICE: 'bastet' packaging is maintained in the 'Git' version control system at:
https://salsa.debian.org/games-team/bastet.git
Please use:
git clone https://salsa.debian.org/games-team/bastet.git
to retrieve the latest (possibly unreleased) updates to the package.
Need to get 43.5 kB of source archives.
Get:1 http://archive.ubuntu.com/ubuntu noble/universe bastet 0.43-7build1 (dsc) [1,995 B]
Get:2 http://archive.ubuntu.com/ubuntu noble/universe bastet 0.43-7build1 (tar) [29.1 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble/universe bastet 0.43-7build1 (diff) [12.4 kB]
Fetched 43.5 kB in 0s (133 kB/s)
dpkg-source: info: extracting bastet in bastet-0.43
dpkg-source: info: unpacking bastet_0.43.orig.tar.gz
dpkg-source: info: unpacking bastet_0.43-7build1.debian.tar.xz
dpkg-source: info: using patch list from debian/patches/series
dpkg-source: info: applying 01_local_hiscores_file_check.patch
dpkg-source: info: applying 02_manpage_global_hiscore_path.patch
dpkg-source: info: applying debian-bug-931550.patch
dpkg-source: info: applying format-security.patch
```

7) В каталоге исходного кода выполнена сборка пакета командой make. Сборка завершилась успешно, сформирован исполняемый файл bastet

```
[16-03-2026 19:23:58] {40ab33fa} student@vm-amd:~/lab3/src-bastet/bastet-0.43$ cd ~/lab3/src-bastet/bastet-0.43
[16-03-2026 19:24:07] {40ab33fa} student@vm-amd:~/lab3/src-bastet/bastet-0.43$ make
g++ -DDEBUG -Wall -c -o Ui.o Ui.cpp
Ui.cpp: In member function 'int Bastet::BorderedWindow::GetMinX()':
Ui.cpp:82:9: warning: variable 'y' set but not used [-Wunused-but-set-variable]
   82 |     int y,x;
      |         ^
Ui.cpp: In member function 'int Bastet::BorderedWindow::GetMaxX()':
Ui.cpp:94:9: warning: variable 'y' set but not used [-Wunused-but-set-variable]
   94 |     int y,x;
      |         ^
Ui.cpp: In function 'Bastet::Dot Bastet::BoundingRect(const std::string&)':
Ui.cpp:174:24: warning: narrowing conversion of '(maxlen + 1)' from 'size_t' {aka 'long unsigned int'} to 'int' [-Wnarrowing]
   174 |     return (Dot)(maxlen+1, splits.size());
      |                        ^
Ui.cpp:174:38: warning: narrowing conversion of 'splits.std::vector<std::__cxx11::basic_string<char>>::size()' from 'std::vector<std::__cxx11::basic_string<char>>::size_type' {aka 'long unsigned int'} to 'int' [-Wnarrowing]
   174 |     return (Dot)(maxlen+1, splits.size());
      |                        ^
Ui.cpp: In member function 'int Bastet::Ui::MenuDialog(const std::vector<std::__cxx11::basic_string<char>>>&)':
Ui.cpp:236:17: warning: narrowing conversion of '(width + 5)' from 'size_t' {aka 'long unsigned int'} to 'int' [-Wnarrowing]
   236 |     Dot d=(width+5, choices.size());
      |               ^
Ui.cpp:236:32: warning: narrowing conversion of '(& choices)->std::vector<std::__cxx11::basic_string<char>>::size()' from 'std::vector<std::__cxx11::basic_string<char>>::size_type' {aka 'long unsigned int'} to 'int' [-Wnarrowing]
   236 |     Dot d=(width+5, choices.size());
      |                        ^
g++ -DDEBUG -Wall -c -o main.o main.cpp
g++ -DDEBUG -Wall -c -o Block.o Block.cpp
g++ -DDEBUG -Wall -c -o Well.o Well.cpp
g++ -DDEBUG -Wall -c -o BlockPosition.o BlockPosition.cpp
g++ -DDEBUG -Wall -c -o Config.o Config.cpp
g++ -DDEBUG -Wall -c -o BlockChooser.o BlockChooser.cpp
g++ -DDEBUG -Wall -c -o BastetBlockChooser.o BastetBlockChooser.cpp
BastetBlockChooser.cpp: In member function 'virtual Bastet::BlockType Bastet::BastetBlockChooser::GetNext(const Bastet::Well*, const Bastet::Queue&)':
BastetBlockChooser.cpp:118:76: warning: 'std::binder2nd<Operation> std::bind2nd(const Operation&, const _Tp&) [with Operation = greater_equal<int>; _Tp = long int]' is deprecated: use 'std::bind' instead [-Wdeprecated-declarations]
   118 |     int pos=find_if(blockPercentages.begin(),blockPercentages.end(),bind2nd(greater_equal<int>(),random()%100)) - blockPercentages.begin();
      |                                                                    ~~~~~^~~~~~
In file included from /usr/include/c++/13/bits/stl_function.h:1438,
                 from /usr/include/c++/13/string:49,
                 from /usr/include/c++/13/bits/locale_classes.h:40,
                 from /usr/include/c++/13/bits/ios_base.h:41,
                 from /usr/include/c++/13/streambuf:43,
                 from /usr/include/c++/13/bits/streambuf_iterator.h:35,
                 from /usr/include/c++/13/iterator:46,
                 from /usr/include/boost/array.hpp:44,
                 from Block.hpp:22,
                 from BlockChooser.hpp:22,
                 from BastetBlockChooser.hpp:22,
                 from BastetBlockChooser.cpp:19:
/usr/include/c++/13/backward/binders.h:172:5: note: declared here
   172 |     bind2nd(const _Operation& __fn, const _Tp& __x)
      |     ^~~~~~
BastetBlockChooser.cpp: In member function 'virtual Bastet::BlockType Bastet::NoPreviewBlockChooser::GetNext(const Bastet::Well*, const Bastet::Queue&)':
BastetBlockChooser.cpp:214:76: warning: 'std::binder2nd<Operation> std::bind2nd(const Operation&, const _Tp&) [with Operation = greater_equal<int>; _Tp = long int]' is deprecated: use 'std::bind' instead [-Wdeprecated-declarations]
   214 |     int pos=find_if(blockPercentages.begin(),blockPercentages.end(),bind2nd(greater_equal<int>(),random()%100)) - blockPercentages.begin();
      |                                                                    ~~~~~^~~~~~
In file included from /usr/include/c++/13/backward/binders.h:172:5: note: declared here
   172 |     bind2nd(const _Operation& __fn, const _Tp& __x)
      |     ^~~~~~
g++ -gdbg -o bastet Ui.o main.o Block.o Well.o BlockPosition.o Config.o BlockChooser.o BastetBlockChooser.o -lncurses -lboost_program_options
```

8) В Makefile добавлена цель install для установки бинарного файла, пакет установлен в систему

```
GNU nano 7.2 Makefile *
SOURCES=Ui.cpp main.cpp Block.cpp Well.cpp BlockPosition.cpp Config.cpp BlockChooser.cpp BastetBlockChooser.cpp
PROGRAM=bastet
LDFLAGS+=-lncurses
CXXFLAGS+=-gdbg -Wall
CXXFLAGS+=-DDEBUG -Wall
CXXFLAGS+=-pg
LDLAGS+=-pg

all: $(PROGRAM)

depend: *.hpp $(SOURCES)
$(CXX) -MM $(SOURCES) > depend

include depend

$(PROGRAM): $(SOURCES:.cpp=.o)
$(CXX) -gdbg -o $(PROGRAM) $(SOURCES:.cpp=.o) $(LDLAGS) -lboost_program_options

clean:
rm -f $(SOURCES:.cpp=.o) $(PROGRAM)

mrproper: clean
rm -f **

install: bastet
install -d /usr/local/bin
install -m 755 bastet /usr/local/bin/bastet
```

```
[16-03-2026 19:24:39] {40ab33fa} student@vm-amd:~/lab3/src-bastet/bastet-0.43$ cd ~/lab3/src-bastet/bastet-0.43
[16-03-2026 19:28:02] {40ab33fa} student@vm-amd:~/lab3/src-bastet/bastet-0.43$ sudo make install
install -d /usr/local/bin
install -m 755 bastet /usr/local/bin/bastet
[16-03-2026 19:28:25] {40ab33fa} student@vm-amd:~/lab3/src-bastet/bastet-0.43$ ls -l /usr/local/bin/bastet
-rwxr-xr-x 1 root root 693008 Mar 16 19:28 /usr/local/bin/bastet
[16-03-2026 19:28:33] {40ab33fa} student@vm-amd:~/lab3/src-bastet/bastet-0.43$ stat -c '%a %n' /usr/local/bin/bastet
755 /usr/local/bin/bastet
```

9) Проверен запуск установленного пакета от другого пользователя командой. Программа успешно запускается в непривилегированном режиме и создаёт пользовательский файл настроек в домашнем каталоге

```
[16-03-2026 19:28:38] {40ab33fa} student@vm-amd:~/lab3/src-bastet/bastet-0.43$ sudo useradd -m lab3test
[16-03-2026 19:31:19] {40ab33fa} student@vm-amd:~/lab3/src-bastet/bastet-0.43$ sudo -u lab3test /usr/local/bin/bastet --help | head -n 5
bastet: using a user-specific high scores file: /home/lab3test/.bastetscores
as the global high scores file /var/games/bastet.scores2 is not writable
bastet: creating a new user-specific high scores file /home/lab3test/.bastetscores
```

10) Создан файл task10.log со списком всех установленных пакетов

```
[[16-03-2026 19:37:17] {40ab33fa} student@vm-amd:~/lab3/src-bastet/bastet-0.43$ dpkg -l > task10.log
[[16-03-2026 19:42:01] {40ab33fa} student@vm-amd:~/lab3/src-bastet/bastet-0.43$ head -n 20 task10.log
Desired=Unknown/Install/Remove/Purge/Hold
| Status=Not/Inst/Conf-files/Unpacked/half-instd/trig-await/Trig-pend
|/ Err?=(none)/Reinst-required (Status,Err: uppercase=bad)
++-----+
|| Name                               Version                               Architecture Description
++-----+
ii adduser                             3.137ubuntu1                         all          add and remove users and groups
ii amd64-microcode                     3.20250311.1ubuntu0.24.04.1         amd64        Processor microcode firmware for AMD CPUs
ii apparmor                            4.0.1really4.0.1-0ubuntu0.24.04.4   amd64        user-space parser utility for AppArmor
ii apport                              2.28.1-0ubuntu3.8                   all          automatically generate crash reports for debugging
ii apport-core-dump-handler            2.28.1-0ubuntu3.8                   all          Kernel core dump handler for Apport
ii apport-symptoms                     0.25                                 all          symptom scripts for apport
ii appstream                           1.0.2-1build6                        amd64        Software component metadata management
ii apt                                 2.8.3                                amd64        commandline package manager
ii apt-utils                           2.8.3                                amd64        package management related utility programs
ii autoconf                            2.71-3                               all          automatic configure script builder
ii automake                            1:1.16.5-1.3ubuntu1                 all          Tool for generating GNU Standards-compliant Makefiles
ii autopoint                           0.21-14ubuntu2                      all          tool for setting up gettext infrastructure in a source package
ii autotools-dev                       20220109.1                           all          Update infrastructure for config.{guess,sub} files
ii base-files                           13ubuntu0.3                         amd64        Debian base system miscellaneous files
ii base-passwd                         3.6.3build1                          amd64        Debian base system master password and group files
```

11) Сформирован файл task11.log со списком пакетов, связанных с установкой и работой компилятора gcc

```
[[16-03-2026 19:42:06] {40ab33fa} student@vm-amd:~/lab3/src-bastet/bastet-0.43$ apt-cache depends gcc | tee task11.log
gcc
Depends: cpp
Depends: cpp-x86-64-linux-gnu
Depends: gcc-13
Depends: gcc-x86-64-linux-gnu
Conflicts: gcc-doc
| Recommends: libc6-dev
Recommends: <libc-dev>
  libc6-dev
Suggests: gcc-multilib
Suggests: make
  make-guile
Suggests: manpages-dev
Suggests: autoconf
Suggests: automake
Suggests: libtool
Suggests: flex
Suggests: bison
Suggests: gdb
Suggests: gcc-doc
[[16-03-2026 19:43:53] {40ab33fa} student@vm-amd:~/lab3/src-bastet/bastet-0.43$ head -n 30 task11.log
gcc
Depends: cpp
Depends: cpp-x86-64-linux-gnu
Depends: gcc-13
Depends: gcc-x86-64-linux-gnu
Conflicts: gcc-doc
| Recommends: libc6-dev
Recommends: <libc-dev>
  libc6-dev
Suggests: gcc-multilib
Suggests: make
  make-guile
Suggests: manpages-dev
Suggests: autoconf
Suggests: automake
Suggests: libtool
Suggests: flex
Suggests: bison
Suggests: gdb
Suggests: gcc-doc
```

12) Создан файл task12.log со списком пакетов, требующих установленный libgpm2, с использованием обратных зависимостей

```
Suggests: gcc-doc
[[16-03-2026 19:43:56] {40ab33fa} student@vm-amd:~/lab3/src-bastet/bastet-0.43$ apt-cache rdepends --installed libgpm2 | tee task12.log
libgpm2
Reverse Depends:
mc
vim
mc
vim
libncursesw6
libncurses6
[[16-03-2026 19:46:33] {40ab33fa} student@vm-amd:~/lab3/src-bastet/bastet-0.43$ head -n 40 task12.log
libgpm2
Reverse Depends:
mc
vim
mc
vim
libncursesw6
libncurses6
```

13) Создан локальный каталог репозитория /root/localrepo и помещены пять разных версий пакета htop

```
[[16-03-2026 20:04:00] {40ab33fa} student@vm-amd:/tmp$ sudo sh -c 'ls -l /root/localrepo/htop_*_amd64.deb'
/root/localrepo/htop_3.2.2-2_amd64.deb
/root/localrepo/htop_3.3.0-4_amd64.deb
/root/localrepo/htop_3.3.0-5_amd64.deb
/root/localrepo/htop_3.4.0-5_amd64.deb
/root/localrepo/htop_3.4.1-5_amd64.deb
```

14) В каталоге локального репозитория /root/localrepo сгенерированы индексные файлы Packages и Packages.gz, а также создан файл Release с описанием репозитория

```
sudo: the -o option may be used to run a command in a specific directory.
[16-03-2026 20:44:32] {40ab33fa} student@vm-amd:/tmp$ sudo sh -c 'cd /root/localrepo && dpkg-scanpackages ./dev/null > Packages'
dpkg-scanpackages: warning: package htop (filename ./htop_3.4.1-5_amd64.deb) is repeat but newer version; used that one and ignored data from ./htop_3.4.0-5_amd64.deb1
dpkg-scanpackages: warning: package htop (filename ./htop_3.2.2-2_amd64.deb) is repeat; ignored that one and using data from ./htop_3.4.1-5_amd64.deb1
dpkg-scanpackages: warning: package htop (filename ./htop_3.3.0-5_amd64.deb) is repeat; ignored that one and using data from ./htop_3.4.1-5_amd64.deb1
dpkg-scanpackages: warning: package htop (filename ./htop_3.3.0-4_amd64.deb) is repeat; ignored that one and using data from ./htop_3.4.1-5_amd64.deb1
dpkg-scanpackages: warning: Packages with multiple instances but no --multiversion specified:
dpkg-scanpackages: warning: ./htop_3.2.2-2_amd64.deb
dpkg-scanpackages: warning: ./htop_3.3.0-4_amd64.deb
dpkg-scanpackages: warning: ./htop_3.3.0-5_amd64.deb
dpkg-scanpackages: warning: ./htop_3.4.0-5_amd64.deb
dpkg-scanpackages: warning: ./htop_3.4.1-5_amd64.deb
dpkg-scanpackages: warning: Packages in archive but missing from override file:
dpkg-scanpackages: warning: htop
dpkg-scanpackages: info: Wrote 1 entries to output Packages file.
[16-03-2026 20:44:04] {40ab33fa} student@vm-amd:/tmp$ sudo gzip -kf /root/localrepo/Packages
[16-03-2026 20:44:11] {40ab33fa} student@vm-amd:/tmp$ sudo tee /root/localrepo/Release >/dev/null <<'E0f'
> Origin: My Local Repo
Label: My Local Repo
Suite: stable
Version: 1.0
Codename: myrepo
Architectures: amd64
Components: main
Description: My local APT repository
E0f
[16-03-2026 20:44:26] {40ab33fa} student@vm-amd:/tmp$ sudo ls -l /root/localrepo/Packages /root/localrepo/Packages.gz /root/localrepo/Release
-rw-r--r-- 1 root root 77 Mar 16 20:44 /root/localrepo/Packages
-rw-r--r-- 1 root root 548 Mar 16 20:44 /root/localrepo/Packages.gz
-rw-r--r-- 1 root root 162 Mar 16 20:44 /root/localrepo/Release
[16-03-2026 20:44:34] {40ab33fa} student@vm-amd:/tmp$ sudo head -n 20 /root/localrepo/Packages
Packages htop
Version: 3.4.1-5
Architecture: amd64
Maintainer: Daniel Lange <DLange@debian.org>
Installed-Size: 424
Depends: libc6 (>= 2.38), libncursesw6 (>= 6), libtinfo6 (>= 6)
Suggests: lm-sensors, lsof, strace
Filename: ./htop_3.4.1-5_amd64.deb
Size: 171412
MD5sum: 34a59e464850f9d513aaebb670bac2f6
SHA1: 2a45c0a0ac314234360412b5ede66992a112a0e6
SHA256: 42725cc15e215e9286bf169fe734fbf47826dae209f319f6d0837ecb950112fbf
Section: utils
Priority: optional
Homepage: https://htop.dev/
Description: interactive processes viewer
 Htop is an ncurses-based process viewer similar to top, but it
 allows one to scroll the list vertically and horizontally to see
 all processes and their full command lines.
.
[16-03-2026 20:44:39] {40ab33fa} student@vm-amd:/tmp$ sudo cat /root/localrepo/Release
Origin: My Local Repo
Label: My Local Repo
Suite: stable
Version: 1.0
Codename: myrepo
Architectures: amd64
```

15) Обновлён кэш apt

```
[16-03-2026 20:53:49] {40ab33fa} student@vm-amd:/tmp$ sudo apt update
[sudo] password for student:
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble-updates/universe Sources [516 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble-updates/restricted Sources [70.0 kB]
Get:7 http://archive.ubuntu.com/ubuntu noble-updates/main Sources [525 kB]
Get:8 http://archive.ubuntu.com/ubuntu noble-updates/multiverse Sources [27.4 kB]
Get:9 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1,837 kB]
Get:10 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [178 kB]
Get:11 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]
Get:12 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1,566 kB]
Get:13 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [386 kB]
Get:14 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:15 http://archive.ubuntu.com/ubuntu noble-backports/universe Sources [15.0 kB]
Get:16 http://archive.ubuntu.com/ubuntu noble-backports/main Sources [5,972 B]
Get:17 http://archive.ubuntu.com/ubuntu noble-backports/multiverse Sources [1,012 B]
Get:18 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [7,364 B]
Get:19 http://archive.ubuntu.com/ubuntu noble-backports/restricted amd64 Components [216 B]
Get:20 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [13.1 kB]
Get:21 http://archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Components [212 B]
Get:22 http://security.ubuntu.com/ubuntu noble-security/restricted Sources [65.4 kB]
Get:23 http://security.ubuntu.com/ubuntu noble-security/multiverse Sources [20.9 kB]
Get:24 http://security.ubuntu.com/ubuntu noble-security/main Sources [220 kB]
Get:25 http://security.ubuntu.com/ubuntu noble-security/universe Sources [343 kB]
Get:26 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1,540 kB]
Get:27 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.5 kB]
Get:28 http://security.ubuntu.com/ubuntu noble-security/main amd64 c-n-f Metadata [10.4 kB]
Get:29 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Components [212 B]
Get:30 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [976 kB]
Get:31 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [74.2 kB]
Get:32 http://security.ubuntu.com/ubuntu noble-security/universe amd64 c-n-f Metadata [20.8 kB]
Get:33 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Components [212 B]
Fetched 9,640 kB in 5s (1,674 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
67 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

16) Выведен список подключённых репозиторийев и их параметры из *.sources, а также краткая сводка по источникам пакетов

```
[16-03-2026 21:01:33] {40ab33fa} student@vm-amd:/tmp$ grep -R --no-filename -e '^(Types|URIs|Suites|Components):' /etc/apt/sources.list.d/*.sources 20/dev/null
Types: deb deb-src
URIs: http://archive.ubuntu.com/ubuntu/
Suites: noble noble-updates noble-backports
Components: main restricted universe multiverse
Types: deb deb-src
URIs: http://security.ubuntu.com/ubuntu/
Suites: noble noble-security
Components: main restricted universe multiverse
[16-03-2026 21:01:05] {40ab33fa} student@vm-amd:/tmp$ apt-get-cache policy
Package files
100 /var/lib/dpkg/status
500 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Packages
release v024.04.0ubuntu,anoble-security,nnoble,lubuntu,cmultiverse,bnand4
origin security.ubuntu.com
500 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages
release v024.04.0ubuntu,anoble-security,nnoble,lubuntu,crestricted,bnand4
origin security.ubuntu.com
500 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages
release v024.04.0ubuntu,anoble-security,nnoble,lubuntu,cmain,bnand4
origin security.ubuntu.com
100 http://archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Packages
release v024.04.0ubuntu,anoble-backports,nnoble,lubuntu,cmultiverse,bnand4
origin archive.ubuntu.com
100 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Packages
release v024.04.0ubuntu,anoble-backports,nnoble,lubuntu,cuniverse,bnand4
origin archive.ubuntu.com
100 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Packages
release v024.04.0ubuntu,anoble-backports,nnoble,lubuntu,cmain,bnand4
origin archive.ubuntu.com
500 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Packages
release v024.04.0ubuntu,anoble-updates,nnoble,lubuntu,cmultiverse,bnand4
origin archive.ubuntu.com
500 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages
release v024.04.0ubuntu,anoble-updates,nnoble,lubuntu,cuniverse,bnand4
origin archive.ubuntu.com
500 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages
release v024.04.0ubuntu,anoble-updates,nnoble,lubuntu,crestricted,bnand4
origin archive.ubuntu.com
500 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages
release v024.04.0ubuntu,anoble-updates,nnoble,lubuntu,cmain,bnand4
origin archive.ubuntu.com
500 http://archive.ubuntu.com/ubuntu noble/multiverse amd64 Packages
release v024.04.0ubuntu,anoble,nnoble,lubuntu,cmultiverse,bnand4
origin archive.ubuntu.com
500 http://archive.ubuntu.com/ubuntu noble/universe amd64 Packages
release v024.04.0ubuntu,anoble,nnoble,lubuntu,cuniverse,bnand4
origin archive.ubuntu.com
500 http://archive.ubuntu.com/ubuntu noble/restricted amd64 Packages
release v024.04.0ubuntu,anoble,nnoble,lubuntu,crestricted,bnand4
origin archive.ubuntu.com
500 http://archive.ubuntu.com/ubuntu noble/main amd64 Packages
release v024.04.0ubuntu,anoble,nnoble,lubuntu,cmain,bnand4
origin archive.ubuntu.com
Pinned packages:

```

17) Сформирован файл task16.log со списком всех доступных версий htop из локального репозитория (/opt/localrepo)

```
3.4.1-5
[[17-03-2026 13:13:16] {40ab33fa} student@vm-amd:/tmp$ cat task16.log
Available htop versions in local repository (/opt/localrepo):
3.2.2-2
3.3.0-4
3.3.0-5
3.4.0-5
3.4.1-5
```

18) Установлена предпоследняя версия пакета htop (3.4.0-5)

```
[[17-03-2026 13:17:58] {40ab33fa} student@vm-amd:/tmp$ sudo dpkg -i /opt/localrepo/htop_3.4.0-5_amd64.deb
(Reading database ... 191099 files and directories currently installed.)
Preparing to unpack .../htop_3.4.0-5_amd64.deb ...
Unpacking htop (3.4.0-5) over (3.3.0-4build1) ...
Setting up htop (3.4.0-5) ...
Processing triggers for mailcap (3.70+nmu1ubuntu1) ...
Processing triggers for man-db (2.12.0-4build2) ...
[[17-03-2026 13:19:06] {40ab33fa} student@vm-amd:/tmp$ htop --version
htop 3.4.0
[[17-03-2026 13:19:10] {40ab33fa} student@vm-amd:/tmp$ dpkg -l htop
Desired=Unknown/Install/Remove/Purge/Hold
| Status=Not/Inst/Conf-files/Unpacked/halF-conf/Half-inst/trig-aWait/Trig-pend
|/ Err?=(none)/Reinst-required (Status,Err: uppercase=bad)
++-+-----+
|| Name          Version      Architecture Description
++-+-----+
ii htop          3.4.0-5      amd64         interactive processes viewer
```

19) Пакет nano скачан из сетевого репозитория и пересобран

```
[[17-03-2026 13:23:11] {40ab33fa} student@vm-amd:~/lab3/nano-rebuild$ dpkg-deb -b pkg nano-newnano.deb
dpkg-deb: building package 'nano' in 'nano-newnano.deb'.
[[17-03-2026 13:23:16] {40ab33fa} student@vm-amd:~/lab3/nano-rebuild$ sudo dpkg -i nano-newnano.deb
(Reading database ... 191100 files and directories currently installed.)
Preparing to unpack nano-newnano.deb ...
Unpacking nano (7.2-2ubuntu0.1) over (7.2-2ubuntu0.1) ...
Setting up nano (7.2-2ubuntu0.1) ...
Processing triggers for install-info (7.1-3build2) ...
Processing triggers for man-db (2.12.0-4build2) ...
[[17-03-2026 13:23:26] {40ab33fa} student@vm-amd:~/lab3/nano-rebuild$ ls -l /usr/bin/newnano
lrwxrwxrwx 1 student student 9 Mar 17 13:23 /usr/bin/newnano -> /bin/nano
[[17-03-2026 13:23:48] {40ab33fa} student@vm-amd:~/lab3/nano-rebuild$ newnano --version | head -n 1
GNU nano, version 7.2
```

20)

- работа с репозиториями
- обновление индексов пакетов
- установка/удаление/обновление пакетов
- разрешение зависимостей и конфликтов версий
- поиск и просмотр информации о пакетах
- работа с бинарными пакетами

```
ubuntu@ubuntu: ~
ubuntu@ubuntu:~$ apt moo
      (__)
      (oo)
    /-----\
   /  |      | \
  *  / \    / \
   ~ ~      ~ ~
... "Have you mooed today?" ...
ubuntu@ubuntu:~$
```